u_core

RISC_V_SingleCycle_Core

i_clk

i_nrst o_wmem

i_instr[31:0]

i_mem_data[31:0]

o_alu[31:0] o_data[31:0]

o_pc[31:0]

u_alu

ALU

o_zero

i_a[31:0]

i_alu_op[4:0]

i_b[31:0]

o_result[31:0] u_branch_addr BranchAddress

i_instruction[31:0]

i_pc[31:0] o_address[31:0]

u_cu

ControlUnit

i_z

o_bimm

o_call o_m2reg

o_wmem

o_wreg

i_func3[2:0]

i_func7[6:0]
i_opcode[6:0]

o_aluc[4:0] o_alui[1:0] o_pcsrc[1:0]

u_instmem

InstructionMemory

i_address[31:0] o_instruction[31:0]

u_imm

ImmediateGenerator

i_immediate_type[1:0]
i_instruction[31:0]

o_immediate[31:0]

u_datamem

DataMemory

i_clk

i_mem_write

i_address[31:0]

i_write_data[31:0] o_read_data[31:0]

u_jal_addr

JALAddress

i_instruction[31:0]

i_pc[31:0] o_address[31:0]

u_jalr_addr JALRAddress

i_instruction[31:0]

i_source_register[31:0] o_address[31:0]

u_mux_aluB Mux2x32

i_s

i_a0[31:0]

i_a1[31:0] o_y[31:0]

u_mux_call Mux2x32

i_s

i_a0[31:0]

i_a1[31:0] o_y[31:0]

u_mux_memres Mux2x32

i_s

i_a0[31:0]

i_a1[31:0] o_y[31:0]

u_pc_mux Mux4x32

i_a0[31:0]

i_a1[31:0]

i_a2[31:0]
i_a3[31:0]

i_s[1:0]

o_y[31:0]

u_pc4 ProgramCounter

i_pc[31:0] o_pc[31:0]

u_regfile

RegisterFile

i_clk_100MHz
i_nrst_async

i_write_en

i_rd[4:0]

i_rs1[4:0]
i_rs2[4:0]
i_write_data[31:0]

o_read_data1[31:0] o_read_data2[31:0]

result_i__2 RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

result_i__3

RTL_MUX

S=5'b01001 I2

S=5'b01010 I3

S=5'b11001 I4
S=5'b11010 I5

S=5'b00010 I6

S=5'b00000 I0[31:0] S=5'b00001 I1[31:0] S=5'b00011;5'b10011 I10[31:0] S=5'b00111;5'b10111 I11[31:0] S=5'b01111;5'b11111 I12[31:0] S=5'b00110;5'b10110 I13[31:0] S=5'b01000 I14[31:0] S=5'b01100;5'b11100 I15[31:0] S=5'b01101 I16[31:0] S=5'b10000 I17[31:0] S=5'b10001 I18[31:0] S=5'b10010 I19[31:0] S=5'b11000 I20[31:0] S=5'b11101 I21[31:0] S=default I22[31:0] S=5'b01011;5'b11011 I7[31:0] S=5'b00100;5'b10100 I8[31:0] S=5'b00101;5'b10101 I9[31:0]

O[31:0]

S[4:0] +

result0_i RTL_ADD

I0[31:0]

I1[31:0] O[31:0] -

result0_i__0 RTL_SUB

I0[31:0]

I1[31:0] O[31:0] =

result0_i__1 RTL_EQO

I0[31:0]

I1[31:0]

!=

result0_i__2 RTL_NEQ O

I0[31:0]

I1[31:0] <

result0_i__3 RTL_LT O

I0[31:0]

I1[31:0] >=

result0_i__4 RTL_GEQ O

I0[31:0]

I1[31:0] result0_i__5 RTL_XOR

I0[31:0]

I1[31:0] O[31:0] result0_i__6 RTL_OR

I0[31:0]

I1[31:0] O[31:0] result0_i__7 RTL_AND

I0[31:0]

I1[31:0] O[31:0] *

result0_i__8 RTL_MULT

I0[31:0]

I1[31:0] O[31:0]

result0_i__9 RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

result0_i__10 RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0] *

result0_i__11 RTL_MULT

I0[31:0]

I1[31:0] O[63:0] *

result0_i__12 RTL_MULT

I0[31:0]

I1[31:0] O[63:0] /result0_i__13 RTL_DIV

I0[31:0]

I1[31:0] O[31:0] %

result0_i__14 RTL_MOD

I0[31:0]

I1[31:0] O[31:0] /result1_i RTL_DIV

I0[31:0]

I1[31:0] O[31:0] -
result1_i__0 RTL_MINUS

I0[31:0] O[31:0]

!=

result1_i__1 RTL_NEQ O

I0[31:0]

I1[31:0] -

result1_i__2 RTL_MINUS

I0[31:0] O[31:0]

!=

result1_i__3 RTL_NEQ O

I0[31:0]
I1[31:0]

result2_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

result2_i__0 RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0] =

result2_i__1 RTL_EQ

I0

I1 O %

result2_i__2 RTL_MOD

I0[31:0]

I1[31:0] O[31:0] -

result3_i RTL_MINUS

I0[31:0] O[31:0] -

result3_i__0 RTL_MINUS

I0[31:0] O[31:0] <

result3_i__1 RTL_LT O

I0[31:0]

I1[31:0] <

result3_i__2 RTL_LT O

I0[31:0]

I1[31:0] >=

result4_i RTL_GEQ O

I0[31:0]
I1[31:0]

>=

result4_i__0 RTL_GEQ O

I0[31:0]
I1[31:0]

instruction_array[0]_i RTL_ROM

A[4:0] O[31:0] +

o_address_i RTL_ADD

I0[31:0]

I1[31:0] O[31:0]

data_memory_reg RTL_RAM

WCLK WE1 RA1[4:0] RO1[31:0] WA1[4:0] WD1[31:0]

i_func3_i RTL_MUX

V=B"00000001", S=3'b000 I0[7:0] V=B"00000010", S=3'b001 I1[7:0] V=B"00000100", S=3'b100 I2[7:0] V=B"00001000", S=3'b110 I3[7:0] V=B"00010000", S=3'b111 I4[7:0] V=B"00100000", S=3'b101 I5[7:0] V=B"01000000", S=3'b010 I6[7:0] V=B"10000000", S=3'b011 I7[7:0] O[7:0] S[2:0]

i_func7_i RTL_MUX

V=B"001", S=7'b0000000 I0[2:0] V=B"010", S=7'b0100000 I1[2:0] V=B"100", S=7'b0000001 I2[2:0] O[2:0] S[6:0]

i_jal_i RTL_ROM

A[6:0] O

i_lui_i RTL_ROM

A[6:0] O

i_lw_i RTL_ROM

A[6:0] O

i_opcode_i RTL_MUX

V=B"0001", S=7'b1100111 I0[3:0] V=B"0010", S=7'b1100011 I1[3:0] V=B"0100", S=7'b0010011 I2[3:0] V=B"1000", S=7'b0110011 I3[3:0] O[3:0] S[6:0]

i_sw_i RTL_ROM

A[6:0] O

o_aluc0_i RTL_AND

I0

I1 O

o_aluc0_i__0 RTL_AND

I0

I1 O

o_aluc0_i__1 RTL_AND

I0

I1 O

o_aluc0_i__2 RTL_AND

I0

I1 O

o_aluc0_i__3 RTL_AND

I0

I1 O

o_aluc1_i RTL_OR

I0

I1 O

o_aluc1_i__0 RTL_OR

I0

I1 O

o_aluc1_i__1 RTL_OR

I0

I1 O

o_aluc1_i__2 RTL_OR

I0

I1 O

o_aluc1_i__3 RTL_OR

I0

I1 O

o_aluc2_i RTL_OR

I0

I1 O

o_aluc2_i__0 RTL_OR

I0

I1 O

o_aluc2_i__1 RTL_AND

I0

I1 O

o_aluc2_i__2 RTL_OR

I0

I1 O

o_aluc2_i__3 RTL_OR

I0

I1 O

o_aluc2_i__4 RTL_OR

I0

I1 O o_aluc2_i__5 RTL_OR

I0

I1 O

o_aluc3_i RTL_OR

I0

I1 O

o_aluc3_i__0 RTL_AND

I0

I1 O

o_aluc3_i__1 RTL_OR

I0

I1 O

o_aluc3_i__2 RTL_AND

I0

I1 O

o_aluc3_i__3 RTL_AND

I0

I1 O

o_aluc3_i__4 RTL_OR

I0

I1 O

o_aluc3_i__5 RTL_OR

I0

I1 O

o_aluc3_i__6 RTL_OR

I0

I1 O

o_aluc3_i__7 RTL_OR

I0

I1 O

o_aluc4_i RTL_AND

I0

I1 O

o_aluc4_i__0 RTL_OR

I0

I1 O

o_aluc4_i__1 RTL_AND

I0

I1 O

o_aluc4_i__2 RTL_AND

I0

I1 O

o_aluc4_i__3 RTL_OR

I0

I1 O

o_aluc4_i__4 RTL_OR

I0

I1 O

o_aluc4_i__5 RTL_OR

I0

I1 O

o_aluc4_i__6 RTL_OR

I0

I1 O

o_aluc4_i__7 RTL_AND

I0

I1 O

o_aluc4_i__8 RTL_OR

I0

I1 O

o_aluc5_i RTL_OR

I0

I1 O

o_aluc5_i__0 RTL_AND

I0

I1 O

o_aluc5_i__1 RTL_OR

I0

I1 O

o_aluc5_i__2 RTL_AND

I0

I1 O

o_aluc5_i__3 RTL_OR

I0

I1 O

o_aluc5_i__4 RTL_OR

I0

I1 O

o_aluc5_i__5 RTL_AND

I0

I1 O

o_aluc5_i__6 RTL_AND

I0

I1 O

o_aluc5_i__7 RTL_OR

I0

I1 O

o_aluc6_i RTL_AND

I0

I1 O

o_aluc6_i__0 RTL_OR

I0

I1 O

o_aluc6_i__1 RTL_AND

I0

I1 O

o_aluc6_i__2 RTL_AND

I0

I1 O

o_aluc6_i__3 RTL_OR

I0

I1 O

o_aluc6_i__4 RTL_OR

I0

I1 O

o_aluc6_i__5 RTL_OR

I0

I1 O

o_aluc6_i__6 RTL_AND

I0

I1 O

o_aluc6_i__7 RTL_AND

I0

I1 O

o_aluc6_i__8 RTL_OR

I0

I1 O

o_aluc7_i RTL_AND

I0

I1 O

o_aluc7_i__0 RTL_AND

I0

I1 O

o_aluc7_i__1 RTL_OR

I0

I1 O

o_aluc7_i__2 RTL_AND

I0

I1 O

o_aluc7_i__3 RTL_AND

I0

I1 O

o_aluc7_i__4 RTL_OR

I0

I1 O

o_aluc7_i__5 RTL_OR

I0

I1 O

o_aluc7_i__6 RTL_OR

I0

I1 O

o_aluc7_i__7 RTL_OR

I0

I1 O

o_aluc8_i RTL_OR

I0

I1 O

o_aluc8_i__0 RTL_AND

I0

I1 O

o_aluc8_i__1 RTL_OR

I0

I1 O

o_aluc8_i__2 RTL_AND

I0

I1 O

o_aluc8_i__3 RTL_OR

I0

I1 O

o_aluc8_i__4 RTL_OR

I0

I1 O

o_aluc8_i__5 RTL_OR

I0

I1 O

o_aluc9_i RTL_OR

I0

I1 O

o_aluc9_i__0 RTL_AND

I0

I1 O

o_aluc9_i__1 RTL_AND

I0

I1 O

o_aluc9_i__2 RTL_AND

I0

I1 O

o_aluc9_i__3 RTL_OR

I0

I1 O

o_aluc9_i__4 RTL_OR

I0

I1 O

o_aluc9_i__5 RTL_OR

I0

I1 O

o_aluc10_i RTL_OR

I0

I1 O

o_aluc10_i__0 RTL_AND

I0

I1 O

o_aluc10_i__1 RTL_AND

I0

I1 O

o_aluc10_i__2 RTL_OR

I0

I1 O

o_aluc11_i RTL_OR

I0

I1 O

o_aluc11_i__0 RTL_AND

I0

I1 O

o_aluc11_i__1 RTL_OR

I0

I1 O

o_aluc12_i RTL_AND

I0

I1 O

o_aluc12_i__0 RTL_AND

I0

I1 O

o_aluc12_i__1 RTL_OR

I0

I1 O

o_aluc13_i RTL_OR

I0

I1 O

o_aluc14_i RTL_AND

I0

I1 O

o_alui0_i RTL_OR

I0

I1 O

o_alui0_i__0 RTL_OR

I0

I1 O

o_alui1_i RTL_OR

I0

I1 O

o_alui2_i RTL_OR

I0

I1 O

o_bimm0_i RTL_OR

I0

I1 O

o_bimm1_i RTL_OR

I0

I1 O

o_bimm2_i RTL_OR

I0

I1 O

o_bimm3_i RTL_OR

I0

I1 O

o_bimm4_i RTL_OR

I0

I1 O

o_bimm5_i RTL_OR

I0

I1 O

o_bimm6_i RTL_OR

I0

I1 O

o_bimm7_i RTL_OR

I0

I1 O

o_bimm8_i RTL_OR

I0

I1 O

o_bimm9_i RTL_AND

I0

I1 O

o_pcsrc0_i RTL_OR

I0

I1 O

o_pcsrc1_i RTL_OR

I0

I1 O

o_pcsrc1_i__0 RTL_AND

I0

I1 O

o_pcsrc2_i RTL_AND

I0

I1 O =

o_pcsrc2_i__0 RTL_EQ

I0

I1 O

o_wreg0_i RTL_OR

I0

I1 O

o_wreg1_i RTL_OR

I0

I1 O

o_wreg2_i RTL_OR

I0

I1 O

o_wreg3_i RTL_OR

I0

I1 O
o_wreg4_i RTL_OR

I0

I1 O

o_wreg5_i RTL_OR

I0

I1 O

o_wreg6_i RTL_OR

I0

I1 O

o_wreg7_i RTL_OR

I0

I1 O

o_wreg8_i RTL_OR

I0

I1 O

o_wreg9_i RTL_OR

I0

I1 O

o_wreg10_i RTL_OR

I0

I1 O

o_wreg11_i RTL_OR

I0

I1 O

o_wreg12_i RTL_OR

I0

I1 O

o_wreg13_i RTL_OR

I0

I1 O

o_wreg13_i__0 RTL_AND

I0

I1 O

o_wreg14_i RTL_OR

I0

I1 O

o_wreg15_i RTL_OR

I0

I1 O
o_wreg16_i RTL_OR

I0

I1 O

o_wreg17_i RTL_OR

I0

I1 O

o_wreg18_i RTL_OR

I0

I1 O

o_wreg19_i RTL_OR

I0

I1 O

o_wreg20_i RTL_OR

I0

I1 O

o_wreg21_i RTL_OR

I0

I1 O

o_wreg22_i RTL_OR

I0

I1 O

register_bank_reg[31]_i RTL_ROM

A[4:0] O

imm_i RTL_MUX

S=2'b00 I0[31:0] S=2'b01 I1[31:0] S=2'b10 I2[31:0] S=2'b11 I3[31:0] O[31:0] S[1:0]

register_bank_reg[31][31:0] RTL_REG_ASYNC C

CECLR D Q +

o_address_i RTL_ADD

I0[31:0]

I1[31:0] O[31:0]

register_bank_reg[30]_i RTL_ROM

A[4:0] O +

target_i RTL_ADD

I0[31:0]

I1[31:0] O[31:0]

register_bank_reg[30][31:0] RTL_REG_ASYNC C

CECLR D Q

o_y_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

register_bank_reg[29]_i RTL_ROM

A[4:0] O

o_y_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

register_bank_reg[29][31:0] RTL_REG_ASYNC C

CECLR D Q

o_y_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

register_bank_reg[28]_i RTL_ROM

A[4:0] O

o_y_i RTL_MUX

S=2'b10 I0[31:0] S=default I1[31:0] O[31:0] S[1:0] o_y_i__0 RTL_MUX

S=2'b01 I0[31:0] S=default I1[31:0] O[31:0] S[1:0] o_y_i__1 RTL_MUX

S=2'b00 I0[31:0] S=default I1[31:0] O[31:0] S[1:0]

u_pc_reg DFF32

i_clk

i_nrst

i_d[31:0] o_q[31:0]

register_bank_reg[28][31:0] RTL_REG_ASYNC C

CECLR D Q +

plusOp_i RTL_ADD

I0[31:0] V=B"100" I1[2:0] O[31:0]

register_bank_reg[27]_i RTL_ROM

A[4:0] O

o_read_data1_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0] o_read_data2_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

!=

o_read_data11_i RTL_NEQ O

I0[4:0]
I1[4:0]

!=

o_read_data21_i RTL_NEQ O

I0[4:0]

I1[4:0]

register_bank[0]_i__0

RTL_MUX

S=5'b00000 I0[31:0] S=5'b00001 I1[31:0] S=5'b01010 I10[31:0] S=5'b01011 I11[31:0] S=5'b01100 I12[31:0] S=5'b01101 I13[31:0] S=5'b01110 I14[31:0] S=5'b01111 I15[31:0] S=5'b10000 I16[31:0] S=5'b10001 I17[31:0] S=5'b10010 I18[31:0] S=5'b10011 I19[31:0] S=5'b00010 I2[31:0] S=5'b10100 I20[31:0] S=5'b10101 I21[31:0] S=5'b10110 I22[31:0] S=5'b10111 I23[31:0] S=5'b11000 I24[31:0] S=5'b11001 I25[31:0] S=5'b11010 I26[31:0] S=5'b11011 I27[31:0] S=5'b11100 I28[31:0] S=5'b11101 I29[31:0] S=5'b00011 I3[31:0] S=5'b11110 I30[31:0] S=5'b11111 I31[31:0] S=5'b00100 I4[31:0] S=5'b00101 I5[31:0] S=5'b00110 I6[31:0] S=5'b00111 I7[31:0] S=5'b01000 I8[31:0] S=5'b01001 I9[31:0]

O[31:0]

S[4:0] register_bank[0]_i__1

RTL_MUX

S=5'b00000 I0[31:0] S=5'b00001 I1[31:0] S=5'b01010 I10[31:0] S=5'b01011 I11[31:0] S=5'b01100 I12[31:0] S=5'b01101 I13[31:0] S=5'b01110 I14[31:0] S=5'b01111 I15[31:0] S=5'b10000 I16[31:0] S=5'b10001 I17[31:0] S=5'b10010 I18[31:0] S=5'b10011 I19[31:0] S=5'b00010 I2[31:0] S=5'b10100 I20[31:0] S=5'b10101 I21[31:0] S=5'b10110 I22[31:0] S=5'b10111 I23[31:0] S=5'b11000 I24[31:0] S=5'b11001 I25[31:0] S=5'b11010 I26[31:0] S=5'b11011 I27[31:0] S=5'b11100 I28[31:0] S=5'b11101 I29[31:0] S=5'b00011 I3[31:0] S=5'b11110 I30[31:0] S=5'b11111 I31[31:0] S=5'b00100 I4[31:0] S=5'b00101 I5[31:0] S=5'b00110 I6[31:0] S=5'b00111 I7[31:0] S=5'b01000 I8[31:0] S=5'b01001 I9[31:0]

O[31:0]

S[4:0]

register_bank[0]1_i RTL_AND

I0

I1 O

!=

register_bank[0]2_i RTL_NEQ O

I0[4:0]
I1[4:0]

register_bank[31:0]_i RTL_MUX

S=1'b1 I0

S=default I1 O S

register_bank_reg[0][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[0]_i RTL_ROM

A[4:0] O

register_bank_reg[1][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[1]_i RTL_ROM

A[4:0] O

register_bank_reg[2][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[2]_i RTL_ROM

A[4:0] O

register_bank_reg[3][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[3]_i RTL_ROM

A[4:0] O

register_bank_reg[4][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[4]_i RTL_ROM

A[4:0] O

register_bank_reg[5][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[5]_i RTL_ROM

A[4:0] O

register_bank_reg[6][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[6]_i RTL_ROM

A[4:0] O

register_bank_reg[7][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[7]_i RTL_ROM

A[4:0] O

register_bank_reg[8][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[8]_i RTL_ROM

A[4:0] O

register_bank_reg[9][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[9]_i RTL_ROM

A[4:0] O

register_bank_reg[10][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[10]_i RTL_ROM

A[4:0] O

register_bank_reg[11][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[11]_i RTL_ROM

A[4:0] O

register_bank_reg[12][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[12]_i RTL_ROM

A[4:0] O

register_bank_reg[13][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[13]_i RTL_ROM

A[4:0] O

register_bank_reg[14][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[14]_i RTL_ROM

A[4:0] O

register_bank_reg[15][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[15]_i RTL_ROM

A[4:0] O

register_bank_reg[16][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[16]_i RTL_ROM

A[4:0] O

register_bank_reg[17][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[17]_i RTL_ROM

A[4:0] O

register_bank_reg[18][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[18]_i RTL_ROM

A[4:0] O

register_bank_reg[19][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[19]_i RTL_ROM

A[4:0] O

register_bank_reg[20][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[20]_i RTL_ROM

A[4:0] O

register_bank_reg[21][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[21]_i RTL_ROM

A[4:0] O

register_bank_reg[22][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[22]_i RTL_ROM

A[4:0] O

register_bank_reg[23][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[23]_i RTL_ROM

A[4:0] O

register_bank_reg[24][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[24]_i RTL_ROM

A[4:0] O

register_bank_reg[25][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[25]_i RTL_ROM

A[4:0] O

register_bank_reg[26][31:0] RTL_REG_ASYNC C

CECLR D Q

register_bank_reg[26]_i RTL_ROM

A[4:0] O

register_bank_reg[27][31:0] RTL_REG_ASYNC C

CECLR D Q

result_i__1 RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

result_i__0 RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

result_i RTL_MUX S

S=1'b1 I0[31:0] S=default I1[31:0] O[31:0]

o_zero_i RTL_ROM

A[31:0] O

>>>SHIFT_RIGHT_i__0 RTL_ARSHIFT

I2

I0[31:0]

I1[4:0] O[31:0]

>>SHIFT_RIGHT_i RTL_RSHIFT

I2

I0[31:0]

I1[4:0] O[31:0] <<SHIFT_LEFT_i RTL_LSHIFT

I2

I0[31:0]

I1[4:0] O[31:0]

o_instruction_i RTL_MUX S
S=1'b1 I0[31:0] S=default I1[31:0] O[31:0] <

o_instruction1_i RTL_LT O

I0[31:0] V=X"00000020" I1[31:0]

o_pc[31:0]

o_mem[31:0]

o_instr[31:0]

o_alu[31:0]

i_nrst

i_clk